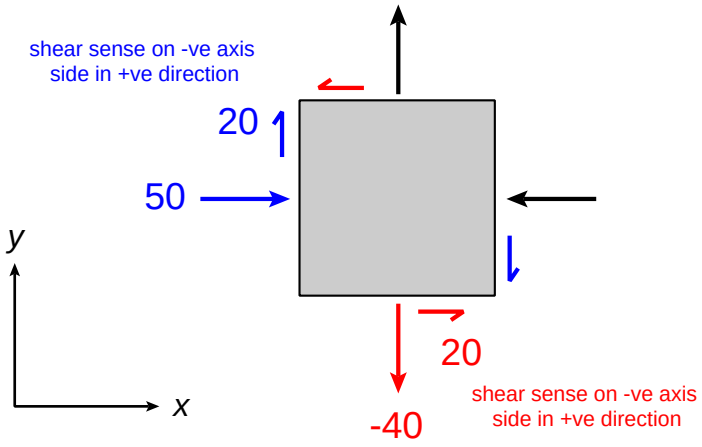
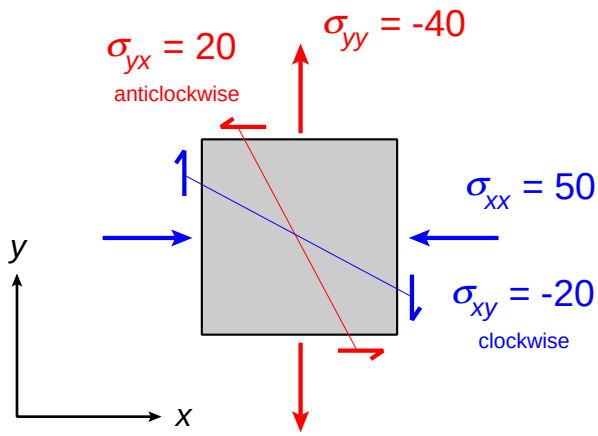
Worked solution for Exercise 1b

$$\begin{bmatrix} 50 & 20 \\ 20 & -40 \end{bmatrix}$$

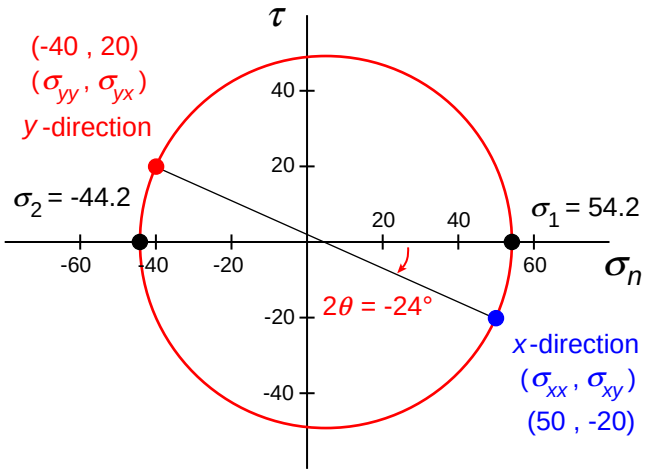
Step (a)

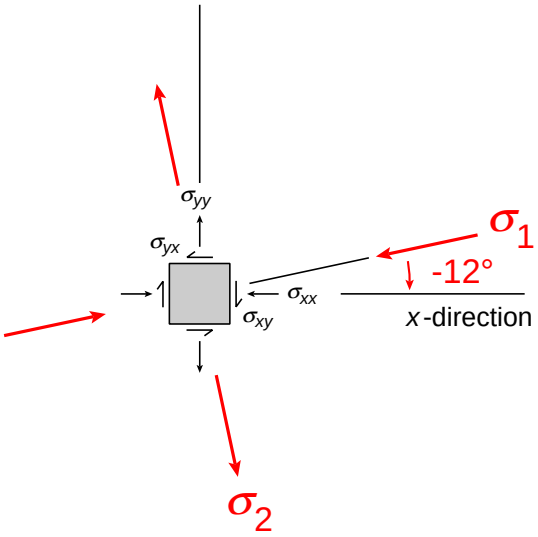
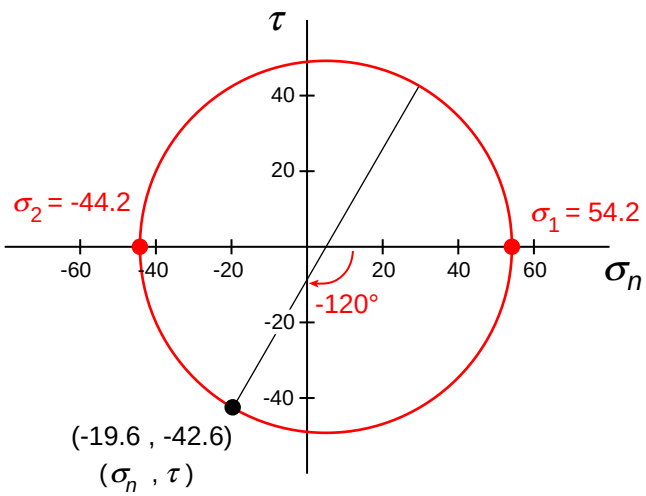
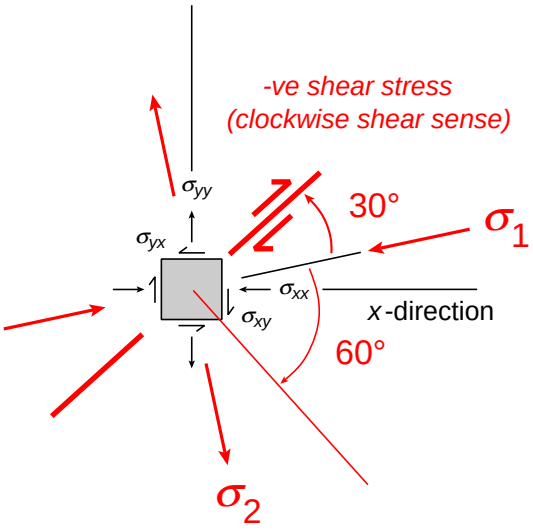


Step (b)



Step (c)





Step (d)